

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC2-068***

“Camden-Crystal II 69 kV”

20 MW Energy, 7.6 MW Capacity

February 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Alamo Solar I, LLC, the Interconnection Customer (IC), has proposed a **20 MW MFO (7.6 MWC)** upgrade to their prior solar queue project AC2-067 “Camden-Crystal II 69 kV” also to be located at 1731 West Consolidated Road in Eaton, Ohio. The projects combined will have a total capability of **69.9 MW** with **26.5 MW** being recognized by PJM as capacity. Both projects will be behind the same Point of Interconnection (POI) and will share the same interconnection facilities. Therefore, PJM studied the total output of both AC2-067 “Camden-Crystal I 69 kV” and AC2-068 “Camden-Crystal II 69 kV” together which is reflected in the Network Impact section of this report. (Also refer to the AC2-067 “Camden-Crystal I 69 kV System Impact Study report.)

The proposed in-service date for this project is **July 1, 2019. This study does not imply a Dayton Power and Light (DP&L) commitment to this in-service date.**

Point of Interconnection

AC2-068 “Camden-Crystal II 69 kV” solar project will utilize the same Point of Interconnection as the Interconnection Customer’s prior queue project, the AC2-067 “Camden-Crystal I 69 kV” solar project. This interconnection is at a tap location on the Hutchings-Crystal 69 kV line between Crystal and Camden Substations. Presently, the Hutchings-Crystal 69kV line serves distribution loads at Camden, Gratis, and Germantown Substations. A single point of interconnection will be constructed to connect both the AC2-067 and AC2-068 Projects to the Dayton System assuming the developer builds a single 69kV lead to the POI on the Crystal-Camden 69kV line section. See Attachment 1 for the One Line Diagram of the proposed interconnection.

Cost Summary

The AC2-067 “Camden-Crystal I 69 kV” project is responsible for the connection facilities to the Dayton Power and Light system. **AC2-068 “Camden-Crystal II 69 kV”** project will share these same interconnection facilities.

The AC2-068 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 0
Non Direct Connection Network Upgrades	\$ 0
Allocation for New System Upgrades	\$ 0
Contribution for Previously Identified Upgrades	\$ 0
Total Costs	\$ 0

The required Attachment Facilities, Direct Connection, and Non-Direct Connection work for the interconnection of the AC2-068 solar project to the DP&L Transmission System is detailed in the following sections. The associated one-line with the generation project is shown in **Attachment 1**.

Attachment Facilities

Attachment Facilities work is addressed by prior queue project AC2-067. No Attachment Facilities are required to be constructed by the Transmission Owner for AC2-068.

Direct Connection Cost Estimate

There is no Direct Connection scope of work required for this project.

Non-Direct Connection Cost Estimate

Non-Direct Connection work is addressed by prior queue project AC2-067. No Non-Direct Connection Facilities are required to be constructed by the Transmission Owner for AC2-068.

Dayton Interconnection Requirements

The Dayton Power and Light Company (DP&L) has prepared this Facilities Connection Requirements document to ensure compliance with North American Electric Reliability Council (NERC) Reliability Standards and applicable Regional Reliability Organization, sub regional, Power Pool, and individual Transmission Owner planning criteria and facility connection requirements in compliance to NERC Standard FAC-001-2. These connection requirements apply to all generation facilities, transmission facilities, and end-users connecting to the DP&L transmission system. Detailed information outlining DP&L interconnection requirements can be reviewed utilizing the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-dayton/dayton-facilities-connection-requirements.ashx>

Schedule

Schedule is contingent upon the AC2-067 “Camden-Crystal I 69 kV” project as defined in their System Impact Study report. Based on the extent of the Dayton Non-Direct Connection and Attachment upgrades required to support the AC2-067 project, it is expected to take a minimum of **12 months** from the date of a fully executed Interconnection Construction Service Agreement to complete the installation subject to market conditions and vendor lead times. This includes the requirement for the Interconnection Customer to make a preliminary payment to Dayton which funds the construction of the Non-Direct Connection facilities and the first three months of construction work in earnest for the Attachment Facility and Direct Connection upgrades. It assumes that there will be no environmental or permitting issues to implement the Non-Direct Connection upgrades for this project and that all system outages will be allowed when requested.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

Dayton Metering Requirements

The Interconnection Customer will be required to comply with all Dayton Revenue Metering Requirements for Generation Interconnection Customers as outlined in the link below. The Revenue Metering Requirements may be found within the Dayton Power & Light Co. "Requirements for the Connection of Facilities to the Dayton Power & Light Co. Transmission System" document located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-dayton/dayton-facilities-connection-requirements.ashx>

<http://www.pjm.com/planning/design-engineering/to-tech-standards.aspx>

Network Impacts

Queue projects AC2-067 & AC2-068 were studied for network impacts as one project. They were evaluated as a total 69.9 MW (Capacity 26.5 MW) injection tapping the Camden – Crystal 69 kV line in the Dayton area. The total of projects AC2-067 and AC2-068 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). The total of projects AC2-067 and AC2-068 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Base Case Used

Summer Peak Analysis – 2020 Case

Contingency Descriptions

The following contingencies resulted in overloads:

None.

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None.

Short Circuit

(Summary of impacted circuit breakers)

None

Affected System Analysis & Mitigation

LGEE Impacts:

None

MISO Impacts:

None

OVEC Impacts:

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Light Load Analysis - 2020

Light Load Studies to be conducted as required by PJM Manual 14B.

Not required for solar projects.

System Reinforcements

Short Circuit

(Summary form of Cost allocation for breakers will be inserted here if any)

None

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

Stability:

Find the finalized AC2-067/AC2-068 Stability study report in **Attachment 3**.

Real-time system operating conditions outside of the stability testing criteria included in this report could occur in this region causing curtailments to maintain stability of the BES system.

Reactive Power Requirement:

The Generation Interconnection Customer shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading to 0.95 lagging measured at the high-side of the facility substation transformers.

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None.

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

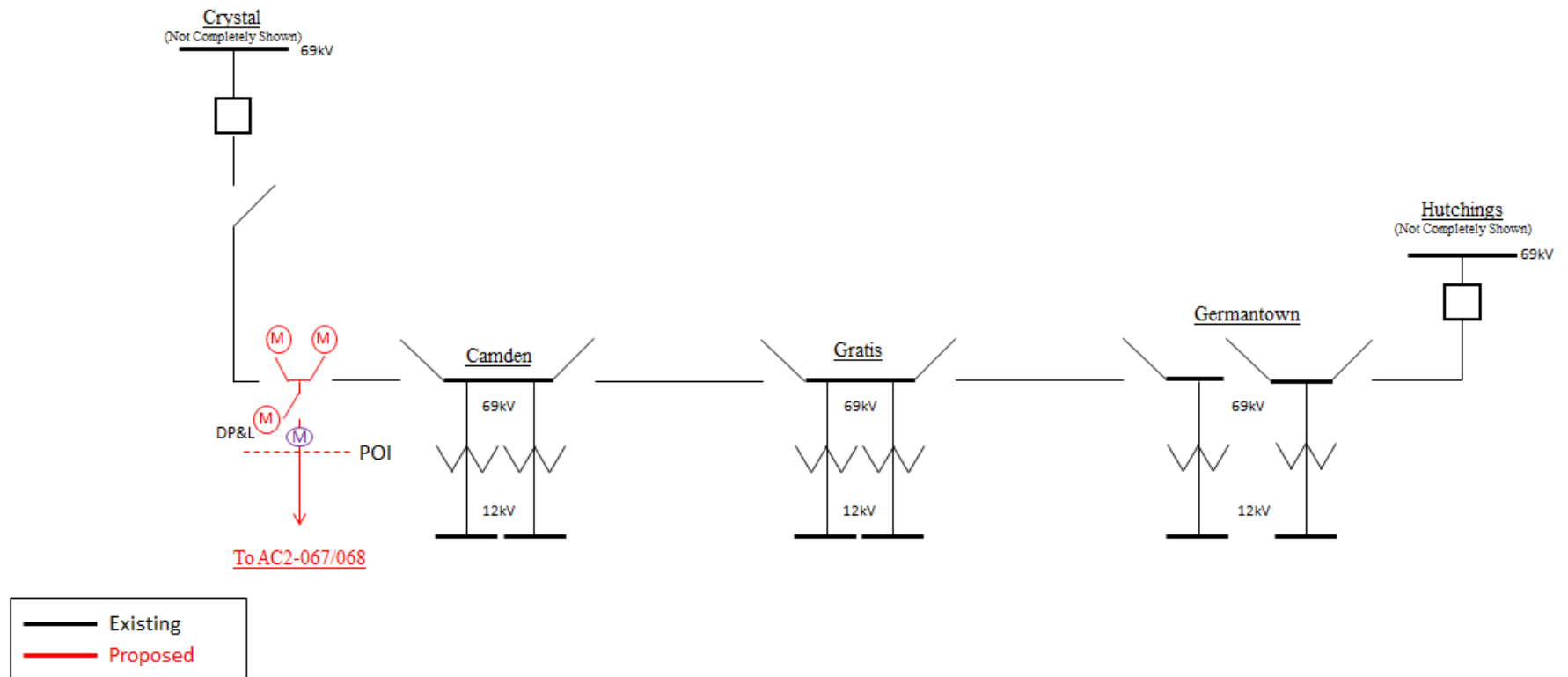
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None.

Attachment 1. One Line Diagram AC2-068 "Camden-Crystal II 69 kV"

AC2-068 Project



Attachment 2. Site Plan
AC2-068 "Camden-Crystal II 69 kV"



Attachment 3. Stability Analysis Report

AC2-067_AC2-068 System Impact Study

Dynamic Simulation Analysis

Version 0

January 16th, 2018

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Executive Summary

Generator Interconnection Request AC2-067 is for a 49.9 MW Maximum Facility Output (MFO) solar photovoltaic generating facility consisting of 24 x 2.09 MW SMA Sunny Central 2200 inverters. AC2-067 has a Point of Interconnection (POI) on a tap of the Camden – Crystal 69 kV line in the Dayton Power and Light system, Preble county, Ohio.

This report describes a dynamic simulation analysis of AC2-067_AC2-068 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2020 summer peak case, modified to include applicable queue projects. AC2-067_AC2-068 has been dispatched online at maximum power output, with approximately unity power factor at the POI.

The AC2-067_AC2-068 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

Relevant High Speed Reclosing (HSR) contingencies were identified in the P1 contingency list table.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the fault contingencies tested on the 2020 summer peak case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AC2-067_AC2-068 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

1. Introduction

Generator Interconnection Request AC2-067 is for a 49.9 MW Maximum Facility Output (MFO) solar photovoltaic generating facility consisting of 24 x 2.09 MW SMA Sunny Central 2200 inverters. AC2-067 has a Point of Interconnection (POI) on a tap of the Camden – Crystal 69 kV line in the Dayton Power and Light system, Preble county, Ohio.

This analysis is effectively a screening study to determine whether the addition of AC2-067_AC2-068 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AC2-067_AC2-068 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

Generator Interconnection Request AC2-067 is for a 49.9 MW Maximum Facility Output (MFO) solar photovoltaic generating facility consisting of 24 x 2.09 MW SMA Sunny Central 2200 inverters. AC2-067 has a Point of Interconnection (POI) on a tap of the Camden – Crystal 69 kV line in the Dayton Power and Light system, Preble county, Ohio.

Generator Interconnection Request AC2-068 is for a 20 MW increase in energy injection capability of the solar photovoltaic generating facility interconnection request AC2-067. AC2-068 consists of 10 x 2.09 MW SMA Sunny Central 2200 inverters.

Figure 1 shows the simplified one-line diagram of the AC2-067 and AC2-068 Queue Projects. Table 1 and Table 2 list the parameters given in the impact study data and the corresponding parameters of the AC2-067 and AC2-068 loadflow model, respectively.

The dynamic model for the AC2-067 and AC2-068 plant is based on the SMA ‘SMASC126’ and ‘SMAPPC18’ PSS/E user models provided by the developer as an attachment to the System Impact Study Data package.

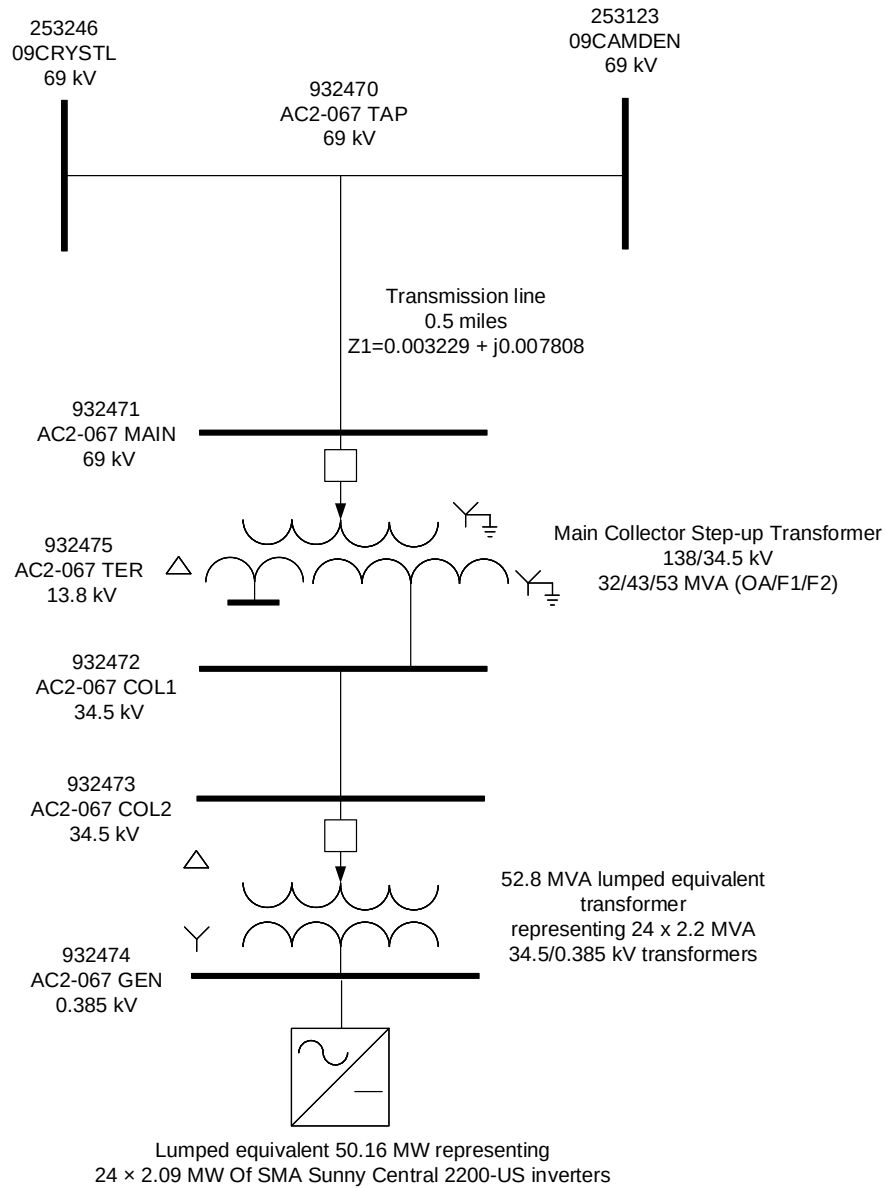


Figure 1: AC2-067 Plant Model

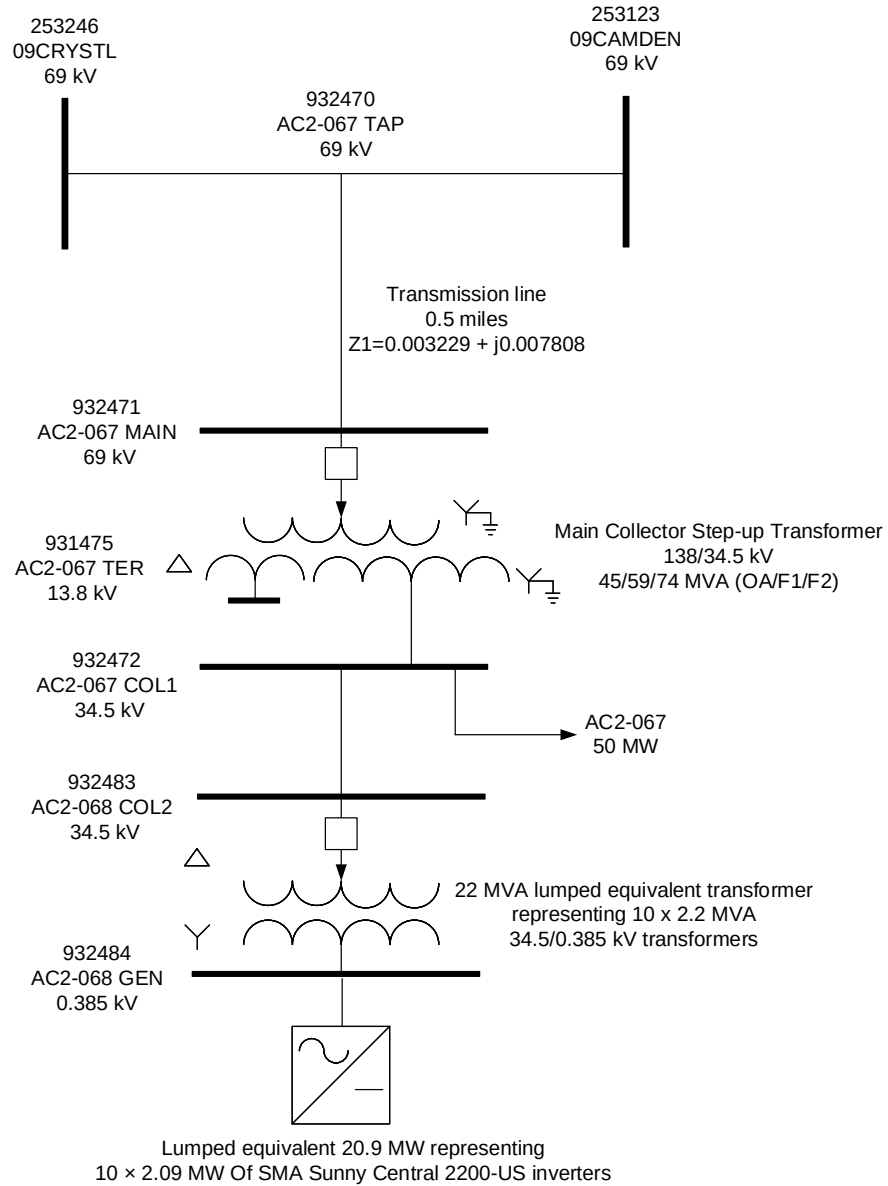


Figure 2: AC2-068 Plant Model

Table 1: AC2-067 Plant Model

	Impact Study Data	Model
Generators	24 x SMA Sunny Central 2200-US 2.09 MW inverters MVA base = 2.2 MVA Vt = 0.385 kV Unsaturated sub-transient reactance = NA	Lumped equivalent representing 24 x Sunny Central 2200-US 2.09 MW inverters Pgen 50.16 MW Pmax 50.16 MW Pmin 0 MW Qmax 16.553 MVA _r Qmin -16.553 MVA _r Mbase 52.8 MVA Zsorce j10000 pu @ Mbase

GSU Transformers	24 x 34.5/0.385 kV two winding transformers Rating = 2.2 MVA (OA) Transformer base = 2.2 MVA Impedances @ MVA base High to Low = $0.007132 + j0.057056$ pu Number of taps = Not given Tap step size = Not given	Lumped equivalent representing 24 x 34.5/0.385 kV two winding transformers Rating = 52.8 MVA (OA) Transformer base = 52.8 MVA Impedances @ MVA base High to Low = $0.007132 + j0.057056$ pu Number of taps = 5 Tap step size = 2.5 %
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Collector Transformer	1 x 69/34.5/13.8 kV three winding transformer Rating = 32/43/53 MVA (OA/F1/F2) Transformer base = 32 MVA Impedances @ MVA base High to Low = $0.003996 + j0.099920$ pu High to Tertiary = $0.006400 + j0.159870$ pu Low to Tertiary = $0.002000 + j0.049960$ pu Number of taps = Not given Tap step size = Not given	1 x 69/34.5/13.8 kV three winding transformer Rating = 32/43/53 MVA (OA/F1/F2) Transformer base = 32 MVA Impedances @ MVA base High to Low = $0.003996 + j0.099920$ pu High to Tertiary = $0.006400 + j0.159870$ pu Low to Tertiary = $0.002000 + j0.049960$ pu Number of taps = 5 Tap step size = 2.5 %
Auxiliary Load	0.0 MW + 0.0 MVAR	Not modeled
Station Load	0.0 MW + 0.0 MVAR	Not modeled
Transmission Line	Length = 0.5 miles Voltage = 69 kV MVA base = 100 Impedance = $0.003229 + j0.007808$ Susceptance = 0.022892	Length = 0.5 miles Voltage = 69 kV MVA base = 100 Impedance = $0.003229 + j0.007808$ Susceptance = 0.022892

	Impact Study Data	Model
Generators	10 x SMA Sunny Central 2200-US 2.09 MW inverters MVA base = 2.2 MVA Vt = 0.385 kV Unsaturated sub-transient reactance = NA	Lumped equivalent representing 10 x Sunny Central 2200-US 2.09 MW inverters Pgen 20.9 MW Pmax 20.9 MW Pmin 0 MW Qmax 6.897 MVAr Qmin -6.897 MVAr Mbase 22 MVA Zsorce j10000 pu @ Mbase
GSU Transformers	10 x 34.5/0.385 kV two winding transformers Rating = 2.2 MVA (OA) Transformer base = 2.2 MVA Impedances @ MVA base High to Low = 0.007132 + j0.057056 pu Number of taps = Not given Tap step size = Not given	Lumped equivalent representing 10 x 34.5/0.385 kV 2.2 MVA transformers Rating = 22 MVA (OA) Transformer base = 22 MVA Impedances @ MVA base High to Low = 0.007132 + j0.057056 pu Number of taps = 5 Tap step size = 2.5 %

Collector Transformer	1 x 69/34.5/13.8 kV three winding transformer Rating = 45/59/74 MVA (OA/F1/F2) Transformer base = 45 MVA Impedances @ MVA base High to Low = $0.003331 + j0.099944$ pu High to Tertiary = $0.005330 + j0.159910$ pu Low to Tertiary = $0.001670 + j0.049970$ pu Number of taps = Not given Tap step size = Not given	1 x 69/34.5/13.8 kV three winding transformer Rating = 45/59/74 MVA (OA/F1/F2) Transformer base = 45 MVA Impedances @ MVA base High to Low = $0.003331 + j0.099944$ pu High to Tertiary = $0.005330 + j0.159910$ pu Low to Tertiary = $0.001670 + j0.049970$ pu Number of taps = 5 Tap step size = 2.5 %
Auxiliary Load	0.0 MW + 0.0 MVAR	Not modeled
Station Load	0.0 MW + 0.0 MVAR	Not modeled
Transmission Line	Length = 0.5 miles Voltage = 69 kV MVA base = 100 Impedance = $0.003229 + j0.007808$ Susceptance = 0.022892	Length = 0.5 miles Voltage = 69 kV MVA base = 100 Impedance = $0.003229 + j0.007808$ Susceptance = 0.022892

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process¹.

The selected load flow scenario is the RTEP 2020 summer peak case with the following modifications:

- a) Addition of all applicable queue projects prior to AC2-067_AC2-068.
- b) Addition of AC2-067_AC2-068 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AC2-067_AC2-068.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.
- e) Sensitivity case which excludes the Crystal – New Lebanon 69kV line as built in the RTEP 2020 case has been studied and the results are consistent with the as built RTEP 2020 case study results.

The AC2-067_AC2-068 initial conditions are listed in Table 2, indicating maximum power output with regulating to unity power factor at the POI bus.

Table 2: AC2-067_AC2-068 machine initial conditions

Bus	Name	Unit	PGEN	QGEN	ETERM	POI Voltage
932474	AC2-067 GEN 0.3850	1	50.16	2.4	1.02	1.0145
932484	AC2-068 GEN 0.3850	1	20.9	2.4	1.02	1.0145

Generation within the vicinity of AC2-067_AC2-068 has been dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of AC2-067_AC2-068 is given in Attachment 3.

¹ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

4. Fault Cases

Tables 3 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above.

5. Evaluation Criteria

This study is focused on AC2-067_AC2-068, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AC2-067_AC2-068 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 3%
- b) The AC2-067_AC2-068 is able to ride through faults (except for faults where protective action trips AC2-067_AC2-068).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 3.

Frequency protection in the model is disabled due to the deficiency of PSSE frequency calculation for inverter based generation facilities.

For the studied fault contingencies tested on the 2020 summer peak case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AC2-067_AC2-068 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. System Reinforcement Requirements

None.

Table 3: Contingency List and Stability Study Results**P0: Stead State**

Fault ID	Duration
P0.00	Steady State 20 sec run

P1: Three Phase Faults with normal clearing

Fault ID	Fault description	Clearing Time Normal & HSR (Cycles)	Results
P1.00	3ph @ AC2-067_AC2-068 Tap, normal clear loss of Crystal – AC2-067 Tap – Camden - Hutchings 69 kV line, trip AC2-067_AC2-068, W2-040	8	Stable
P1.01	3ph @ Crystal – New Lebanon 69kV line, normal clear	8	Stable
P1.02	3ph @ Crystal – Garage Road 69kV line, normal clear	8	Stable
P1.03	3ph @ Hutchings – New Lebanon – Crown 69kV line, normal clear loss of New Lebanon 69kV	8	Stable
P1.04	3ph @ Hutchings 69/138kV TF #7, normal clear	8	Stable
P1.05	3ph @ Hutchings 69/138kV TF #8, normal clear	8	Stable
P1.06	3ph @ Hutchings – Benner 69kV line, normal clear	8	Stable
P1.07	3ph @ Hutchings – Yankee 69kV line, normal clear	8	Stable
P1.08	3ph @ Hutchings – Carrollton 69kV line, normal clear	8	Stable
P1.09	3ph @ Hutchings – Manning 69kV line, normal clear	8	Stable

Fault ID	Fault description	Clearing Time Normal & HSR (Cycles)	Results
P1.10	3ph @ Hutchings unit 7, normal clear	8	Stable
P1.11	3ph @ Hutchings – Crown 138kV, normal clear loss of Crown 138/69kV TF	7	Stable
P1.12	3ph @ Hutchings – College Corner 138kV, normal clear	7	Stable
P1.13	3ph @ Hutchings – Sugarcreek 138kV, normal clear	7	Stable
P1.14	3ph @ Hutchings – Middleboro –Hillsboro 138kV, normal clear	7	Stable

P4: SLG Stuck Breaker (SB) Faults at Backup Clearing

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Results
P4.01	SLG @ Hutchings 69kV bus, normal clear loss of Hutchings 69/138kV TF 7, SB @ Hutchings 69kV, delayed clear loss of Hutchings – Germantown 69kV line section, Hutchings – Benner 69kV line	8/23	Stable
P4.02	SLG @ Hutchings 69kV bus, normal clear loss of Hutchings – Benner 69kV line, SB @ Hutchings 69kV, delayed clear loss of Hutchings – Germantown 69kV line section	8/23	Stable
P4.03	SLG @ Hutchings 138kV bus, normal clear loss of Hutchings – Crown 138kV line, SB @ Hutchings 138kV, delayed clear loss of Hutchings – College Corner 138kV line, Hutchings 138/69kV TF 7	7/21	Stable
P4.04	SLG @ Hutchings 138kV bus, normal clear loss of Hutchings – Crown 138kV line, SB @ Hutchings 138kV, delayed clear loss of Hutchings – Middleboro – Hillsboro 138kV line, Hutchings 138/69kV TF 8	7/21	Stable
P4.05	SLG @ Hutchings 138kV bus, normal clear loss of Hutchings – Sugarcreek 138kV line, SB @ Hutchings 138kV, delayed clear loss of Hutchings – College Corner 138kV line, Hutchings 138/69kV TF 7	7/21	Stable

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Results
P4.06	SLG @ Hutchings 138kV bus, normal clear loss of Hutchings – Sugarcreek 138kV line, SB @ Hutchings 138kV, delayed clear loss of Hutchings – Middleboro – Hillsboro 138kV line, Hutchings 138/69kV TF 8	7/21	Stable
P4.07	SLG @ Crystal – Garage Road 69kV line, normal clear, SB @ Crystal, delayed clear loss of Crystal – New Lebanon 69kV line	8/23	Stable
P4.08	SLG @ Garage Road 69kV bus, normal clear loss of Garage Road 69kV bus, SB 2 @ Garage Road, delayed clear loss of Crystal – Garage Road 69kV line	8/23	Stable

P5: SLG Fault with Delayed Clearing

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Results
P5.01	SLG @ Crystal – New Lebanon 69kV line, relay failure @ Crystal, delayed clear	8/61	Stable
P5.02	SLG @ Crystal – Garage Road 69kV line, relay failure @ Crystal, delayed clear	8/61	Stable
P5.03	SLG @ Hutchings – New Lebanon – Crown 69kV line, relay failure @ Hutchings, delayed clear	8/61	Stable
P5.04	SLG @ Hutchings 69/138kV TF #7, relay failure @ Hutchings, delayed clear	8/61	Stable
P5.05	SLG @ Hutchings 69/138kV TF #8, relay failure @ Hutchings, delayed clear	8/61	Stable
P5.06	SLG @ Hutchings – Benner 69kV line, relay failure @ Hutchings, delayed clear	8/61	Stable
P5.07	SLG @ Hutchings – Yankee 69kV line, relay failure @ Hutchings, delayed clear	8/61	Stable
P5.08	SLG @ Hutchings – Carrollton 69kV line, relay failure @ Hutchings, delayed clear	8/61	Stable
P5.09	SLG @ Hutchings – Manning 69kV line, relay failure @ Hutchings, delayed clear	8/61	Stable
P5.10	SLG @ Hutchings – Crown 138kV, normal clear loss of Crown 138/69kV TF, relay failure @ Hutchings, delayed clear	7/52	Stable
P5.11	SLG @ Hutchings – College Corner 138kV, relay failure @ Hutchings, delayed clear	7/52	Stable
P5.12	SLG @ Hutchings – Sugarcreek 138kV, relay failure @ Hutchings, delayed clear	7/52	Stable

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Results
P5.13	SLG @ Hutchings – Middleboro –Hillsboro 138kV, relay failure @ Hutchings, delayed clear	7/52	Stable

P7: SLG Fault with Common Structure Failure

N/A

Attachment 1. PSS/E Dynamic Model

/*****

*** Project: AC2-068

*** POI: Camden - Crystal 69 kV (Dayton)

*** Fuel: Solar

*** Model: SMA Sunn Central 2200-US

*** Version:

*** Size: 2.09 MW x 10 (20.9 MW)

*** OBJ/DLL/LIB: SMASC_C126_33_IVF111.dll

*** OBJ/DLL/LIB: SMASC_B18_33_IVF111.dll

*** PSSE Version: 33

/*****

932484 'USRMDL' 1 'SMASC126' 1 1 0 73 15 193

1.0 1.0 0.0 1.0 0.0 1.0

1.0 0.0 5.0 0.5 2.0

0.0 0.9 0.5 1.0 0.9 0.9

0.0 0.2 0.05 0.4

0.35 0.35 1 2.0 2.0 2.0 0.1 0.098 0.1 0.098 0.2 0.2

0.01 1.0 20.0 0.5 0.7 0.72 0.0 0.125 0.1

1.2 0.1 1.18 1.0 1.15 2.0 0.88 12.0 0.6 5.0 0.5 3.0

65.0 0.1 64.0 1.0 61.5 3.0 59.3 5.0 57.0 3.0 50.0 0.1

30.0 1.0 10.0 30.0 0.0 1.0 1.0/

932484 'USRMDL' 1 'SMAPPC18' 4 0 4 28 9 21 932472 932483 1 0

3 0.5 0.25 0.04 0.0 0.1 1.0 0.04 0

0 1 0 10 -0.001
0.5 0.2 0.04 0 1 0.2 0.2
0.8 0.91 0.915 1.060 1.055 0.3 1.0/
/
CtlMod KP_PF KI_PF PFXdcrTm PFDB KP_Vol KI_Vol VolXdcrTm VolDB
/
PFNomTot VolNomTot QNomTot VolDroop HybCtlTun
/
KP_P KI_P PXdcrTm PDB PNomTot QCommTm PCommTm
/
PFLim FRTThrLo1 FRTThrLo2 FRTThrHi1 FRTThrHi2 FRTHldTm
Reserved

/*****

*** Project: AC2-068 ends

/*****

/*****

*** Project: AC2-067

/*****

/*****

*** From File: T:\stability\QUEUE_PROJECTS_2020Models_07-26-17\WSS\AC2\AC2-067_DynMdl\AC2-067_Rev0_17-11-17.dyr

/*****

/*****

/** Project: AC2-067

/** POI: Camden - Crystal 69 kV (Dayton)

/** Fuel: Solar

/** Model: SMA Sunn Central 2200-US

/** Version:

/** Size: 2.09 MW x 24 (50.16 MW)

/** OBJ/DLL/LIB: SMASC_C126_33_IVF111.dll

/** OBJ/DLL/LIB: SMASC_B18_33_IVF111.dll

/** PSSE Version: 33

/*****

932474 'USRMDL' 1 'SMASC126' 1 1 0 73 15 193

1.0 1.0 0.0 1.0 0.0 1.0

1.0 0.0 5.0 0.5 2.0

0.0 0.9 0.5 1.0 0.9 0.9

0.0 0.2 0.05 0.4

0.35 0.35 1 2.0 2.0 2.0 0.1 0.098 0.1 0.098 0.2 0.2

0.01 1.0 20.0 0.5 0.7 0.72 0.0 0.125 0.1

1.2 0.1 1.18 1.0 1.15 2.0 0.88 12.0 0.6 5.0 0.5 3.0

65.0 0.1 64.0 1.0 61.5 3.0 59.3 5.0 57.0 3.0 50.0 0.1

30.0 1.0 10.0 30.0 0.0 1.0 1.0/

932474 'USRMDL' 1 'SMAPPC18' 4 0 4 28 9 21 932472 932473 1 0

3 0.5 0.25 0.04 0.0 0.1 1.0 0.04 0

0 1 0 10 -0.001

0.5 0.2 0.04 0 1 0.2 0.2

0.8 0.91 0.915 1.060 1.055 0.3 1.0/

/ CtlMod KP_PF KI_PF PFXdcrTm PFDB KP_Vol KI_Vol VolXdcrTm VolDB
/
/ PFNomTot VolNomTot QNomTot VolDroop HybCtlTun
/
/ KP_P KI_P PXdcrTm PDB PNomTot QCommTm PCommTm
/
Reserved PFLim FRTThrLo1 FRTThrLo2 FRTThrHi1 FRTThrHi2 FRTHldTm

/*****

*** Project: AC2-067 ends

/*****

Attachment 2. PSS/E Case Dispatch

Bus Number	Bus Name	Id	In Service	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)
253028	09GRNVIL 69.000	1	1	51.4	51.4	20	-20	20	-20
253028	09GRNVIL 69.000	2	1	50.5	50.5	20	-20	20	-20
253028	09GRNVIL 69.000	3	1	50.5	50.5	20	-20	20	-20
253028	09GRNVIL 69.000	4	1	50.5	50.5	20	-20	20	-20
253060	09OHH W. 69.000	7	1	27.4	27.4	0	-14.06	23	-19.4
253097	09YANKEE 69.000	1	1	16	16	13.92	-3	10	-3
253097	09YANKEE 69.000	2	1	16.5	16.5	14.36	-3	10	-3
253097	09YANKEE 69.000	3	1	16	16	13.92	-3	10	-3
253097	09YANKEE 69.000	4	1	13	13	9.57	-5	11	-5
253097	09YANKEE 69.000	5	1	13	13	0	-6	11	-6
253097	09YANKEE 69.000	6	1	13	13	11.31	-6	11	-6
253097	09YANKEE 69.000	7	1	13	13	9.57	-6	11	-6
253222	09TAIT E 69.000	1	1	87	87	41	-16.4	50.7	-16.4
253222	09TAIT E 69.000	2	1	89	89	41	-16.8	42.8	-16.8
253222	09TAIT E 69.000	4	1	79	79	41	-18.8	50.6	-18.8
253222	09TAIT E 69.000	6	1	80	80	41	-18	50	-18
253223	09TAIT W 69.000	3	1	80	80	41	-14	42	-14
253223	09TAIT W 69.000	5	1	80	80	41	-18	50	-18
253223	09TAIT W 69.000	7	1	81	81	41	-16.8	48.4	-16.8
253263	09TAIT D 4.2000	8	1	10	10	0	-1.215	8.8	-5.2
902531	W2-040C 69.000	1	1	7.6	7.6	0	0	0	0
902532	W2-040E 69.000	1	1	12.4	12.4	0	0	0	0
904721	V4-073C 69.000	1	1	0.95	0.95	0	0	0	0
910511	X3-002 C 69.000	1	1	1.28	1.28	0	0	0	0
914372	Y2-111 E 69.000	1	1	12	12	0	5.76	5.76	-3.96

915582	Y3-080 E 69.000	1	1	8	8	0	3.84	3.84	-2.64
932474	AC2-067 GEN 0.3850	1	1	50.16	50.16	0	2.455	16.55	-16.55
932484	AC2-068 GEN 0.3850	1	1	20.9	20.9	0	2.455	6.897	-6.897